

A PROJECT TITLE
ON
Player Statistics Visualization Using PCA & UMAP

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

in
A.Y.: 2026-27

by

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Department Of Computer Science and Engineering

Name of Title: Player Statistics Visualization Using PCA & UMAP

1. Abstract:

Sports organizations generate large volumes of player performance data, including statistics such as goals, assists, matches played, passing accuracy, shots, and player ratings. Analyzing these high-dimensional datasets using traditional methods can be challenging, making it difficult to identify similarities, performance trends, and player groupings.

This project, "Player Statistics Visualization Using PCA and UMAP," aims to simplify complex player statistics by applying two dimensionality reduction techniques: Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP). These techniques transform multiple statistical features into a two-dimensional representation while preserving important information from the original data.

The reduced data will be visualized using scatter plots, where players with similar performance characteristics appear closer together. This visualization enables easier exploration of player relationships, identification of clusters, and comparison of playing styles. The project also compares the effectiveness of PCA and UMAP in representing player statistics.

The outcome of this project will provide an intuitive and efficient way to analyse player performance data, demonstrating how machine learning techniques can support sports analytics and assist coaches, analysts, and teams in making informed decisions.

2. INTRODUCTION:

Sports analytics has become increasingly important for evaluating player performance and supporting strategic decision-making. Modern sports generate large amounts of statistical data, including metrics such as goals, assists, passes, shots, matches played, and

player ratings. Although these statistics provide valuable insights, analyzing high-dimensional data using traditional methods can be difficult and time-consuming.

Machine learning offers effective techniques to simplify and visualize such complex datasets. In this project, Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) are used to reduce multiple player performance features into a two-dimensional space while preserving meaningful information. The resulting visualizations help identify similarities, clusters, and performance patterns among players.

The primary objective of this project is to compare the visualization capabilities of PCA and UMAP and demonstrate how dimensionality reduction techniques can improve the analysis and interpretation of player statistics. This project highlights the practical application of machine learning in sports analytics and data visualization.

3. SOFTWARE REQUIREMENTS:

The following software is required to develop the Player Statistics Visualization Using PCA and UMAP project:

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.x
- IDE/Editor: Jupyter Notebook, Google Colab, or Visual Studio Code (VS Code)
- Libraries:
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn (sklearn)
 - UMAP (umap-learn)
- Dataset: Sports Player Statistics Dataset (CSV format)



Signature of the Guide



Course Instructor